UNIT-3

1. Explain about conventional & Non-conventional sources in detail.

A.. Conventional Resources:

Conventional energy resources are old, traditional sources of energy that have been used for many decades. They are generally non-renewable - once used they cannot be replaced quickly.

Eg:- coal, Petroleum, Natural gas, firewood

Characteristics:

* They are found deep underground
* And formed over millions of years
* High energy output, suitable for industries
* cause pollution (CO_2, SO_2, smoke)
* Limited availability.

Advantages:

* Provide large, continous energy supply
* Technology for extraction and use is well-developed.
* cheap to produce

Dis advantages:

* Non-renewable → will exhaust in future
* cause air pollution, global warming
* Mining causes land degradation
* Expensive & dangerous.

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2. Non-conventional Energy

Non-conventional resources are modern, alternative energy resources.

They are renewable & eco-friendly, and can be used repeatedly without getting exhausted

eg:- solar energy, wind energy, Biomass, Hydro electric power, wave energy.

Characteristics:

* They are unlimited & naturally replenished
* Do not cause pollution
* Available locally.
* High initial cost, but cheap to maintain

Advantages:

* Renewable - never run out
* Environment - friendly
* Reduce dependence on fossil fuels
* Create rural employment.

Dis Advantages:

* Require high initial investment
* some are weather-dependent
* Need large area

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2. Draw & explain the operation of Hydel power generation systems.

Principle:

The working principle of a Hydel power plant is based on the conversion of the potential energy of water into electrical energy.

Main components of Hydel power plant:

1. Dam:

A dam is constructed across a river to store a large quantity of water & create a reservoir.

2. Reservoir:

The stored water behind the dam has potential energy due to its height.

3. Penstock:

It is a large concrete pipe which carriers water from the reservoir to the turbine under high pressure

4. Turbine:

The high pressure water from the penstock rotates the blades of turbine. And it converts the kinetic energy of flowing water into mechanical energy.

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5. Generator:

As the turbine rotates generator also rotates & converts the mechanical energy into electrical energy

6. Transformer:

The electricity in the transformer will be step up the voltage to transmit over large distance

7. Powerhouse:

It protects the generator, transformer from weather & ensures smooth operation

Energy Conversion:

P.E → K.E → M.E → Electrical energy

Advantages:

1. It needs low initial investment, no fuel charges.
2. It needs less time to commission the plant
3. Its efficiency is more compared to thermal power plant.

DisAdvantages:

1. It involves high capital cast due to construction of dam.
2. skilled & experience hands are requir to build the plant.

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3. It requires high cost of transmission lines

Diagram:

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3. Explain the operation of thermal power Generation systems with layout diagram

A. Principle:

It works on the principle of conversion of heat energy into electrical energy.

It operates on the Rankine cycle

Main components of thermal power plant

1. Coal Handling point.

The coal brought from the mines is stored in the coal storage yard and it is sent to the boiler furnace using belt conveyors.

2. Boiler:

The boiler burns coal to produce heat energy. It converts water into steam in the boiler tubes.

3. Superheater:

The superheater heats the steam which passed through the boiler to increase its temp. and efficiency before entering the turbine.

4. Turbine:

The superheated steam from the boiler is sent to turbine. The high pressure steam makes the turbine blades to rotate

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5. Generator:

When the turbine rotates, the generator converts mechanical energy into electrical energy.

6. Condenser:

The exhaust steam from the turbine is condensed into water using cold water from the cooling tower.

7. Cooling Tower:

The hot water from the condenser is cooled in the cooling tower & reused in the condenser.

8. Water Pump:

It pumps the condensed water back to the boiler for steam generation.

9. Chimney:

The flue gases produced in the boiler are released into atmosphere through a tall chimney.

Energy conversion:

chemical energy → Heat energy → M.E → Electrical E

Advantages:

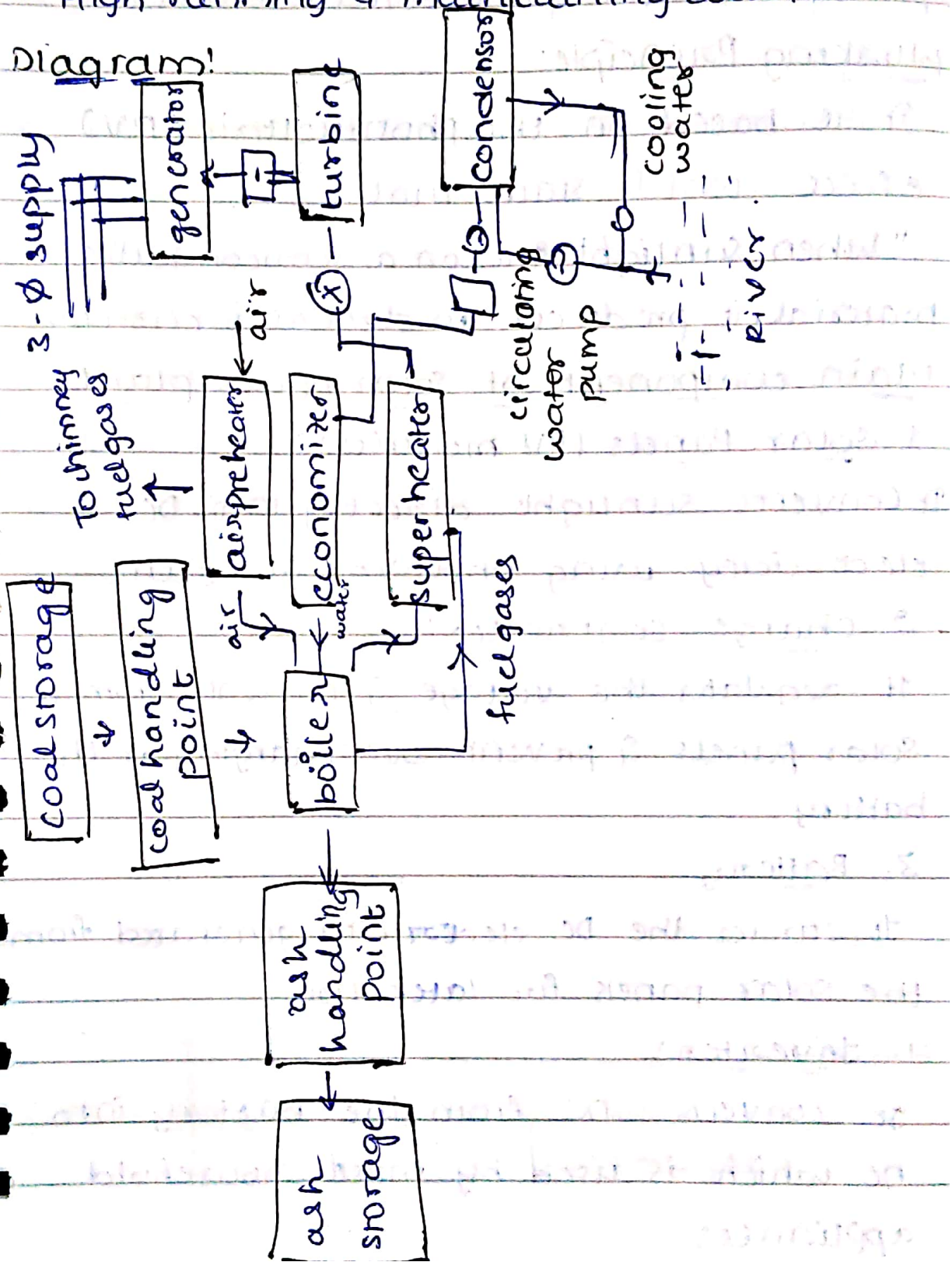
* Can be built near fuel sources
* Easy to control power output
* works continously & provides a stable power supply.

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Disadvantages:

* Its efficiency is low nearly 25-1. to 27-1. compared to hydel power plant
* Transport of fuel is major problem
* Air pollution is a major difficulty.
* High running & maintaining costs.

Diagram:



4. Explain solar & Wind power generation systems layouts.

A. A solar power generator is a system that converts solar energy into electrical energy using solar panels. It works on the principle of the photovoltaic effect.

Working Principle:

It is based on the photovoltaic (PV) effect which states that

"When sunlight falls on a semiconductor material, it produces an electric current."

Main components of solar Power plant:

1. Solar Panels (PV modules):

It converts sunlight directly into DC electricity using photovoltaic cells.

2. Charge Controller:

It regulates the voltage & current from the solar panels & prevents overcharging of the battery.

3. Battery

It stores the DC electricity generated from the solar panels for later use

4. Inverter:

It converts DC from the battery into AC which is used by most household appliances.

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5. load:

The electrical devices powered by the generator.

Energy conversion:

Solar Energy → Electrical E (DC) → A.C.

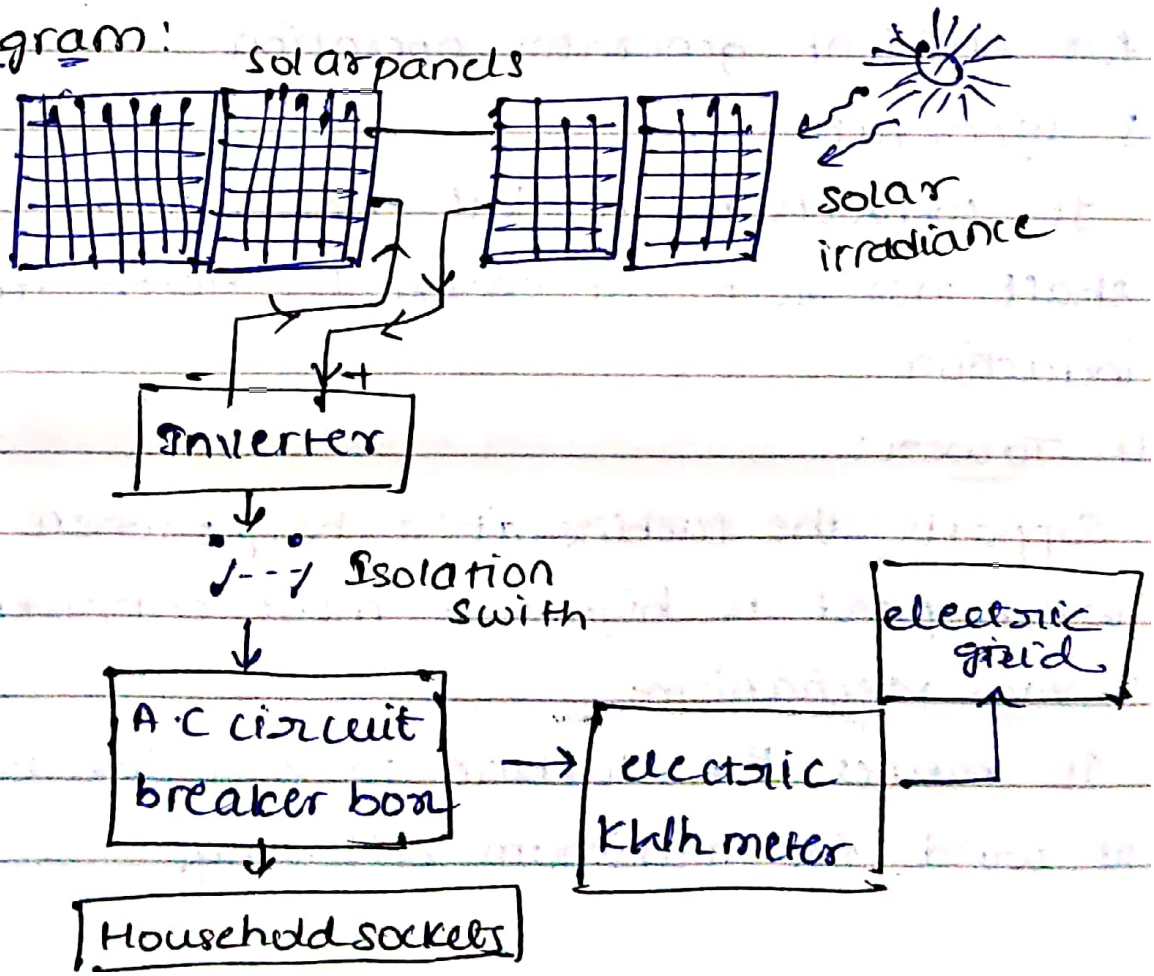
Advantages:

* Renewable & clean source of energy
* low maintainance & operating cost
* Works even in remote areas.
* Reduces electricity bills.

Dis-Advantages:

* cannot generate electricity at night.
* Depends on sunlight availability
* High initial installation cost.

Diagram:



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Wind power plant:

Principle:

A wind power plant converts the kinetic energy of moving air into electrical energy using wind turbines.

Main components of Wind power plant:

1. Wind turbine:

It consists of blades, hub & nacelle. The blades are designed aerodynamically to catch the wind. When wind blows, blades rotate and drive the rotor.

2. Gearbox:

It increases the rotational speed from the slow-moving rotor to the high speed needed for efficient generator operation.

3. Generator:

It converts mechanical energy of the rotating shaft into electrical energy by electromagnetic induction.

4. Tower:

Supports the turbine at a height where wind speed is higher & more consistent.

5. Yaw Mechanism:

It rotates the turbine to face the direction of wind for maximum efficiency.

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6. Controller & Brake System:

The controller starts or stops the turbine depending on wind speed. The Brake system stops the motor during very high winds for safety.

7. Transformer:

It converts the generated electricity to a higher voltage for transmission.

Energy conversion:

K.E → M.E ⟶ Electrical energy.

Advantages:

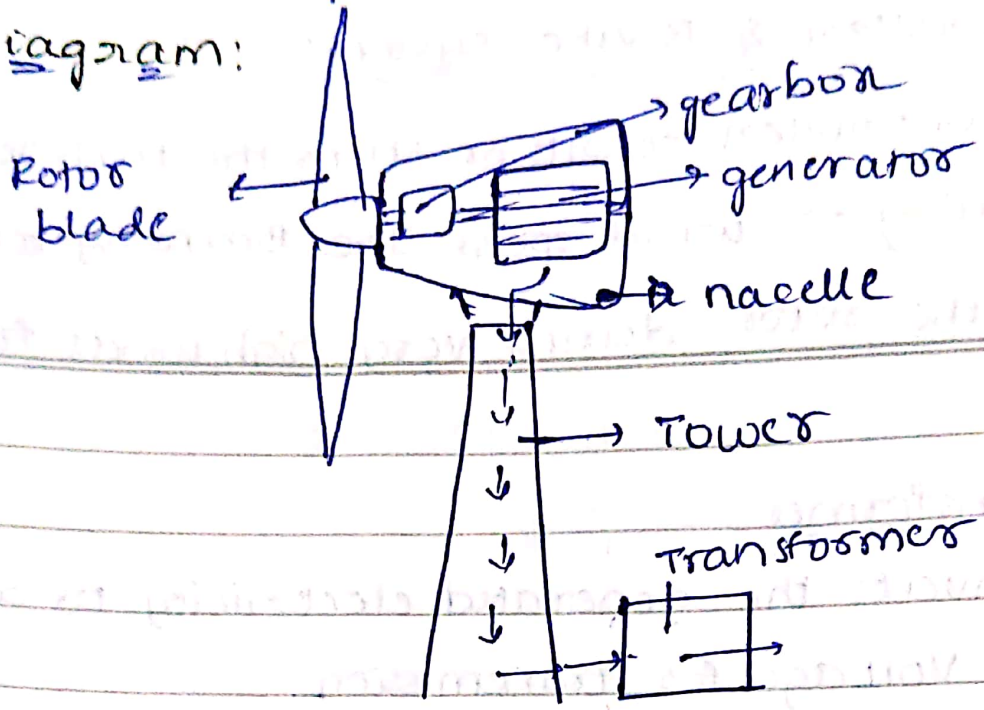
* It is clean & renewable source of energy.
* No fuel cost
* Environment - friendly.
* Suitable for coastal & high-altitude regions.

Dis-Advantages:

* Wind is not constant everywhere
* Requires large open areas
* Noise & visual pollution
* High initial installation cost.

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Diagram:



5. Differentiate between fuse & MCB

A. Fuse	MCB
1. Fuse is a one-time protective device that melts when excessive current flows	1. MCB is an automatic switching device that trips during overload or short circuit.
2. It must be replaced after every fault.	2. It can be reset manually after the fault is cleared.
3. Slow operation compared to MCB	3. Faster and more reliable operation
4. Consists of a metallic wire which melts	4. Consists of electromagnetic & thermal tripping mechanisms
5. Low cost & simple construction	5. More expensive than fuses due to advanced design

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6. Provides less accurate protection because melting points changes over time.	6. Provides accurate & consistent protection due to calibrated tripping.
7. Cannot be used to manually switch ON/OFF a circuit	7. Can be used as a switch to turn on circuit.
8. Used in small appliances & older installations	8. Used in modern home and industrial distri-bution boards.
9. Breaking capacity is lower	9. Breaking capacity is higher.
10. No indication of the fault	10. Has a trip position indicating fault.

6. Explain how to calculate electricity bill for domestic appliances with a small example

S. Appliance	Power	hrs/days	no. of	energy
1. Tubelight	40W	5hrs	30	6
2. Fan	75W	10hrs	30	225
3. TV	100W	4hrs	30	12
4. Fridge	200W	24hrs	30	144
5. WashingM.	500W	1hr	30	15

E = P x t (Whr) ~~= 1000Kwh = 1~~

~~= P x t / 1000~~ 1000 Watth = 1 Kwh

Total kWh energy consumed = 199.5 kWh

if cost of 1 unit = 6/- & fixed charge of 50 rupees

⇒ (199.5 × 6) + 50 = 1247/-

7. Define earthing also explain the purpose of earthing

A. Earthing is the method of connecting the metallic body of electrical equipment and the neutral point of the supply system to the earth through a thick conducting wire or electrode.

→ It provides a low-resistance path for fault current so that any leakage or excess current is safely discharged into the ground.

Purpose of earthing:

1. Protection from electric shock:

If the metal body of an appliance becomes live due to insulation failure, earthing diverts the fault current into the ground.

→ This prevents electric shock when someone touches the appliance.

2. Protection of electrical equipment:

Faults like short circuits or leakage

currents can damage electrical appliances
→ Earthing provides a safe discharge path & prevents overheating, burning & equipment failure

3. Ensures Proper Operation of Protective Devices.

Fuses, MCB's, and RCCB's operate correctly only when sufficient fault current flows.
→ Earthing helps these devices trip quickly during faults, thereby isolating the faulty circuit.

4. Provides a low-Resistance path for faulty current.

Earthing makes sure that any stray current immediately flows to the ground through the path of least resistance.
→ This prevents dangerous voltage build-up on equipment.

5. Prevents Fire Hazards:

leakage currents may cause overheating, sparking, or insulation burning.
→ Effective earthing reduces such risks & prevents electrical fires

6. Safety during fault conditions:

Earthing ensures that in case of a short

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circuit, the excessive current quickly flows to earth, minimizing the time the system remains in a dangerous condition.

Conclusion:

Earthing is an essential part of any electrical installation. By providing a low-resistance path to ground, it protects people, equipment, buildings, and the entire electrical system from dangerous fault currents, shocks, fires and voltage instability.

Q8) What do you mean by electric shock. What are the safety precautions taken to avoid shock.

An electric shock occurs when electric current passes through the human body causing psychological effects, such as pain, burns, muscle contraction, and even death, depending on the current's magnitude & duration.

Safety precautions to avoid electric shock:

1. Proper earthing of equipment:

Earthing provides a low-resistance path for leakage current and prevents the metal body of appliances from becoming live

2. Use of insulated wires & cables:
Double insulated & high-quality wires prevent accidental contact with live conductors.

3. Installation of Protective devices:
These devices quickly disconnect the supply during overload, short circuit or leakage current, protecting users from shock.

4. Avoid using wet-hands or standing on wet floors
Water reduces body resistance & increases current flow, making electric shock more severe.

5. Do not Overload sockets
Overloading causes overheating, insulation melting, and chances of shock or fire.

6. Turn off Power Before Repairing Appliances:
Always disconnecting the power supply before touching internal parts of an electrical device.

7. Use Proper PPE
Electricians should use rubber gloves, insulated tools & safety shoes to avoid conduction of current.

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8. Use correct Voltage & Rated Appliances

Operating appliances at their rated voltage avoids overheating, which can lead to shock hazards.

9. Regular Maintenance of wiring systems:

Damaged insulation, loose connections, or exposed wires should be repaired immediately to prevent accidental contact.

10. Education & Awareness.

People should be trained on safe handling of electrical equipment, first aid for electric shock, and emergency procedures

Conclusion:

Electric shock is a dangerous condition caused by current passing through the human body, Electrical safety is essential for protecting life, equipment, and property.

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